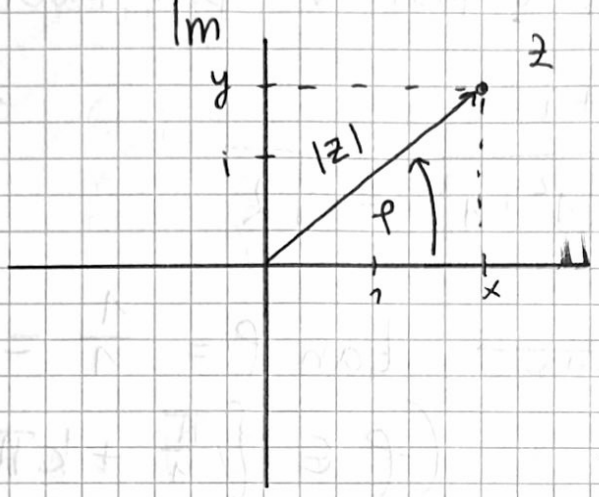
OMA 2

$\mathbb{C}$



$$\tan\varphi = \frac{y}{x}$$

Kartezični zapis super za $+-$

$$z = x + iy$$

kart. x, y koord.

Polarni zapis super za $\cdot, /, \sqrt{}$

$$z = r(\cos\varphi + i\sin\varphi) = r \cdot e^{i\varphi}$$

polarni r, φ koord.

$$\Rightarrow x = r\cdot\cos\varphi \in \mathbb{R}$$

$$y = r\cdot\sin\varphi \in \mathbb{R}$$

$$\Rightarrow r = |z| = \sqrt{x^2+y^2} \in [0,\infty)$$

φ je do $\pm 2k\pi$; $k\in\mathbb{Z}$

ni enolično določeno

$$\varphi = \arctan\frac{y}{x} \in \mathbb{R}$$

Pri določanju upoštevamo kvadrant

φ ... argument z

<u>Primer 1:</u> Zapiši v pol. koord.

$z_1 = 1 + i$

$r = \sqrt{1^2 + 1^2} = \sqrt{2}$

~~$\varphi = \arctan$~~ $\tan\varphi = \frac{1}{1} = 1$

$(\varphi \in \{\frac{\pi}{4} + k\pi\};\ k \in \mathbb{Z}$

$\Rightarrow$ ker smo v 1. kvadrantu

je $\varphi = 45° = \varphi = \frac{\pi}{4}$

$z_2 = -1 - i$

$r = \sqrt{1 + 1} = \sqrt{2}$

$\tan\varphi = \frac{-1}{-1} = 1$

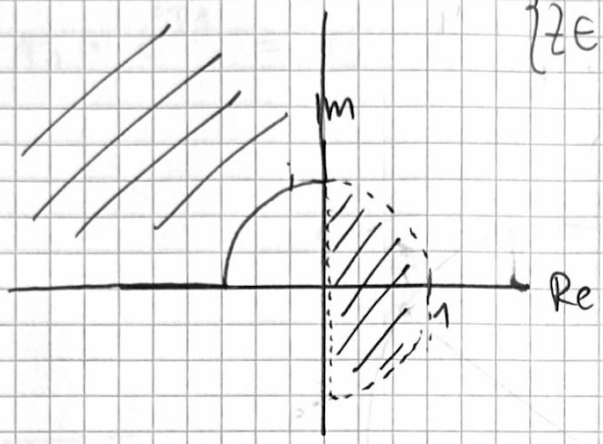
((

ker smo v 3. kvadrantu

$\varphi = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$

Primer 2: Nariši $\{z \in \mathbb{C};\ |z| < 1;\ \mathrm{Re}\, z > 0\}$

$\{z \in \mathbb{C};\ \varphi \in [\frac{\pi}{2}, \pi];\ r \geq 1\}$



MNOŽENJE V POLARNEM ZAPISU:

$$z_i = r_i(\cos\varphi_i + i\sin\varphi_i)$$

$$z_1 \cdot z_2 = r_1(\cos\varphi_1 + i\sin\varphi_1) \cdot r_2(\cos\varphi_2 + i\sin\varphi_2)$$

$$r_1 \cdot r_2(\cos\varphi_1\cos\varphi_2 - \sin\varphi_1\sin\varphi_2 + i(\cos\varphi_1\sin\varphi_2 + \sin\varphi_1\cos\varphi_2))$$

$$= \underbrace{r_1 \cdot r_2}_{\text{produkt abs. vrednosti}} \underbrace{(\cos(\varphi_1+\varphi_2) + i\sin(\varphi_1+\varphi_2))}_{\text{vsota kotov}}$$

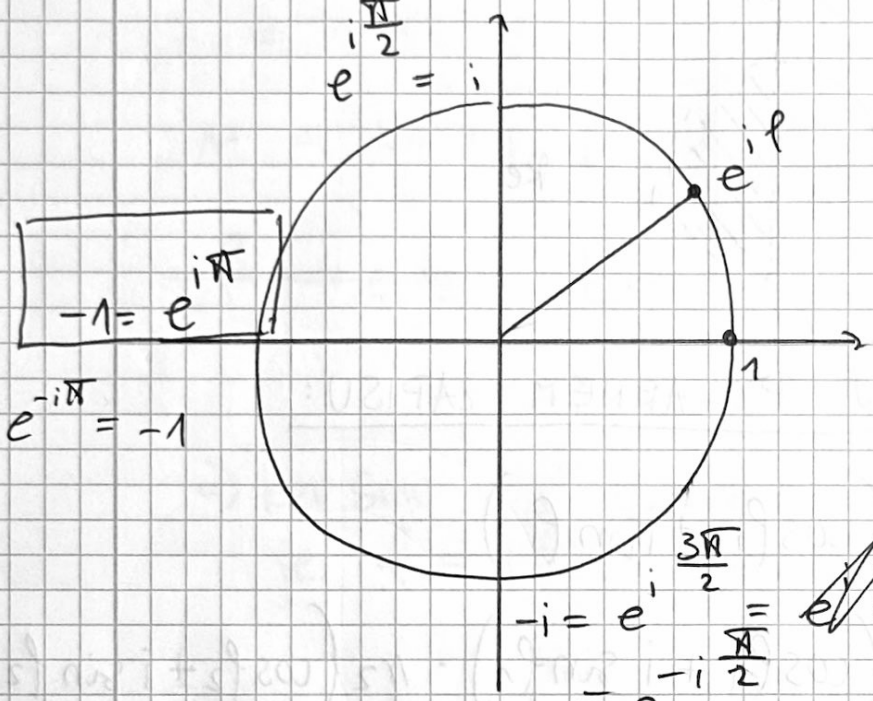
EULERJEVA FORMULA:

$$e^{i\varphi} = \cos\varphi + i\sin\varphi \quad \left.\begin{matrix}\\ \\ \end{matrix}\right\} \text{bližnjica}$$

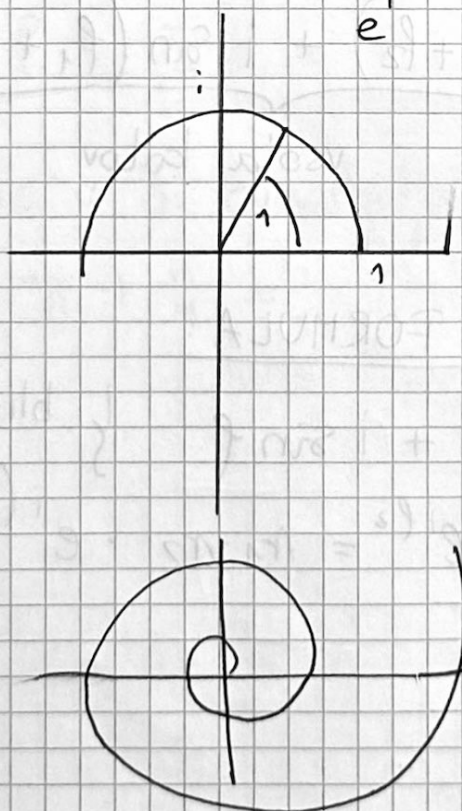
$$r_1 \cdot e^{i\varphi_1} \cdot r_2 e^{i\varphi_2} = r_1 \cdot r_2 \cdot e^{i(\varphi_1+\varphi_2)}$$

$$i = e^{i\frac{\pi}{2}}$$

$$1+i = \sqrt{2}\cdot e^{i\frac{\pi}{4}}$$



Naloga: Nariši $\{re^{ir};\ r \geq 0\}$



torej

1. e^{i1}
2. e^{i2}
3. e^{i3}

$\pi \cdot e^{i\pi} = -\pi$

IGRANJE V POLARNEM ZAPISU:

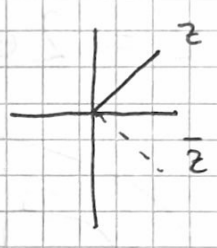
- $r_1 \cdot e^{i\varphi_1} = r_2 \cdot e^{i\varphi_2} \Rightarrow r_1 = r_2 \neq 0$ ~~in $\varphi_1 = \varphi_2$~~

  in $\varphi_1 - \varphi_2 \in 2k\pi$; $k \in \mathbb{Z}$

  ali $r_1 = r_2 = 0$

  je vseeno kakšni so koti

- $\overline{r \cdot e^{i\varphi}} = r \cdot e^{-i\varphi}$



- $(r \cdot e^{i\varphi})^n = r^n \cdot e^{in\varphi}$ ← De Moivre-ova formula

- $(r e^{i\varphi})^{-1} = r^{-1} \cdot e^{-i\varphi}$

  $\uparrow r \neq 0$

- $\frac{r_1 e^{i\varphi_1}}{r_2 e^{i\varphi_2}} = \frac{r_1}{r_2} e^{i(\varphi_1 - \varphi_2)}$

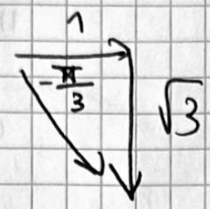
  $\uparrow r_2 \neq 0$

Primer 1: $z = (1 - i\sqrt{3})$

Narišite $z, z^2, z^3, z^{-1}, \ldots$

$r = 2$

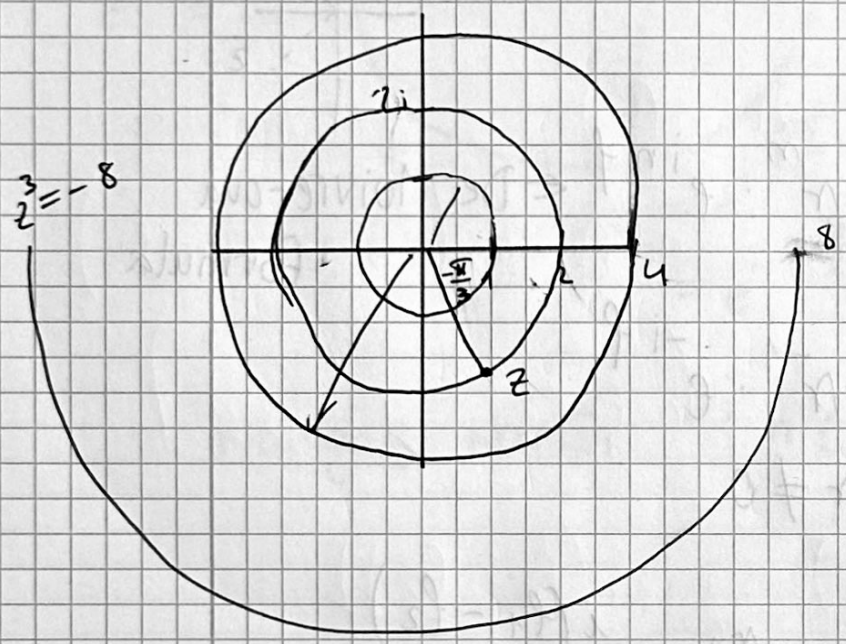
$\varphi = -\frac{\pi}{3}$



$z = 2e^{-i\frac{\pi}{3}}$

$z^2 = 4e^{-i\frac{2\pi}{3}}$

$z^3 = 8e^{-\frac{6\pi}{3}} = -8$



$z^{-1} = \frac{1}{2} \cdot e^{i\frac{\pi}{3}}$

GEOMETRIJA OPERACIJ V $\mathbb{C}$

preslikava: $\mathbb{C} \to \mathbb{C}$ transf. v $\mathbb{C}$

$z \longmapsto z + z_0$ premik za z_0

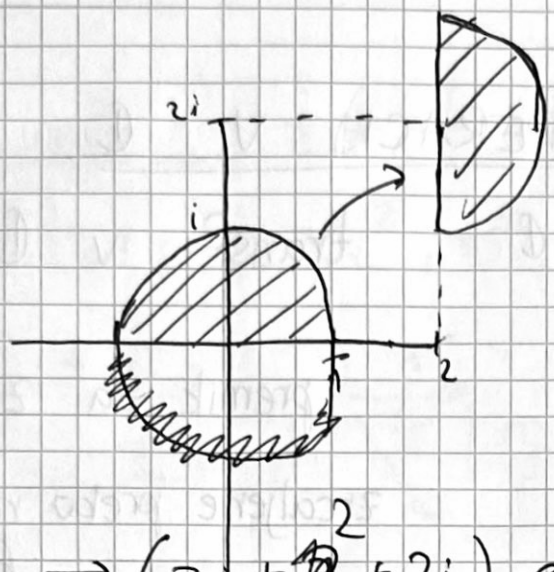
$z \longmapsto \overline{z}$ zrcaljene preko realne osi $Re(z)$

$z \longmapsto r \cdot z ; r \in (0, \infty)$ razteg za r

$z \longmapsto z \cdot e^{i\alpha} ; \alpha \in \mathbb{R}$ rotacija okoli 0 za kot α

$z \longmapsto z \cdot r \cdot e^{i\varphi}$ rotacija za φ in razteg za r
$; r \in (0, \infty)$
$\varphi \in \mathbb{R}$

? DN zrcaljenje preko $Im(z)$



$$z \to (z + 2i)\, e^{-i\frac{\pi}{2}} \quad \times$$

$$z \to z \cdot e^{-i\frac{\pi}{2}} + 2 + 2i$$

primer: $z^5 = 32$

$z = r \cdot e^{i\varphi}$

$z^5 = r^5 \cdot e^{5i\varphi}$

$r^5 e^{5i\varphi} = 2^5 \cdot e^{i \cdot 0}$

$\Rightarrow r = 2$

$\Rightarrow \varphi_1 = 0 = \frac{0}{5}$

$32 = 32 \cdot e^{i \cdot 0}$

$32 = 2^5 \cdot e^{i \cdot 0}$

$= 2^5 \cdot e^{i 2\pi}$

$= 2^5 \cdot e^{i 4\pi}$

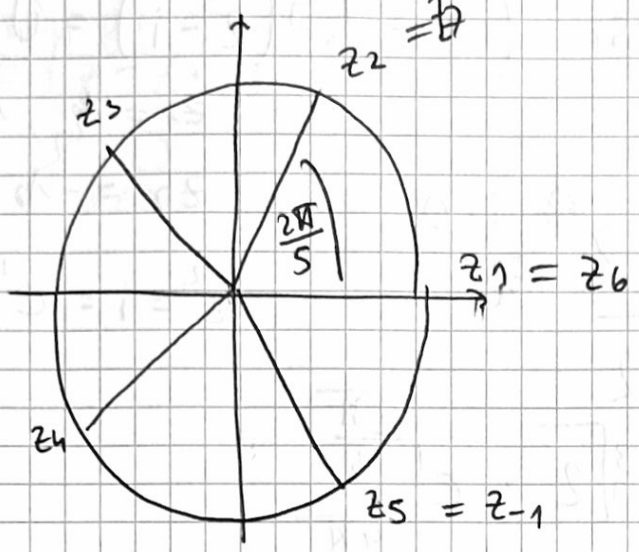
$= 2^5 \cdot e^{-i 2\pi}$

$\varphi_2 = \frac{2\pi}{5}$ $\quad \varphi_{-1} = \frac{-2\pi}{5}$

$\varphi_3 = \frac{4\pi}{5}$

⋮

neskončno možnih kotov



skos je $\frac{2\pi}{5}$

5 rešitev
geometrijsko sestavljajo pravilni 5-kotnik.
Vsota rešitev je 0.

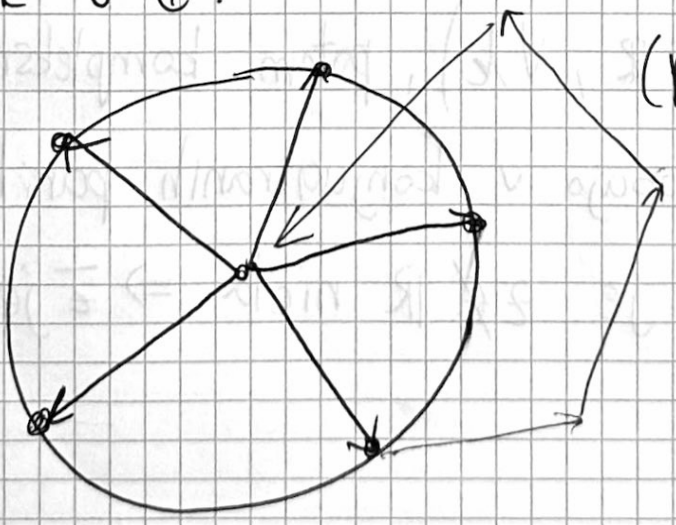
KORENJENJE V $\mathbb{C}$.

Rešitve $z^n = re^{i\varphi}$ $\quad r \neq 0$

$$z_k = \underbrace{\sqrt[n]{r}}_{\in \mathbb{R}} \; e^{i\frac{\varphi + 2k\pi}{n}} \qquad k \in \{0, 1, \ldots, n-1\}$$

DOKAZ

n različnih rešitev sestavljajo pravilni n-kotnik v $\mathbb{C}$.



(pridem nazaj, ker so koti in dolžine isti)

primer: Reši $(z^3-2)(z^2-i)=0$

$z_1 = 1 \cdot e^{i\frac{\pi}{4}}$

$z_2 = -1 \cdot e^{i(\frac{\pi}{4}+\pi)} = -e^{i\frac{\pi}{4}}$

$z^2 = i = e^{i\frac{\pi}{2}}$

$z^3 = 2 = 2 e^{i\cdot 0}$

$z_3 = \sqrt[3]{2}$

$z_4 = \sqrt[3]{2} \cdot e^{i\frac{2\pi}{3}}$

$z_5 = \sqrt[3]{2} \cdot e^{i\frac{4\pi}{3}}$

OSNOVNI IZREK ALGEBRE:

Vsaka enačba $p(z)=0$, kjer je

$p(z) = a_n \cdot z^k + a_{n-1} z^{k-1} + \dots + a_0$; $a_k \in \mathbb{C}$;

$a_n \neq 0$, ima (šteto z v večkratnostjo) natanko n ničel v $\mathbb{C}$. $\forall k \in \{0, \dots, n\}$

Torej: $p(z) = (z-z_1) \cdot \dots \cdot (z-z_n)$, $z_k \in \mathbb{C}$

Dodatno: če so vsi koeficienti realni ($a_k \in \mathbb{R}$, $\forall k$), potem kompleksne ničle nastopajo v konjugiranih parih:

če je $z \notin \mathbb{R}$ ničla $\Rightarrow \overline{z}$ je tudi ničla.

$$p(z) = a_n \underbrace{(z-x_1)(z-x_p)}_{\text{realne ničle}} \cdot \underbrace{Q_1 \cdot Q_2 \cdot \ldots \cdot Q_{\frac{n-p}{2}}}_{\text{kvadratni v } \mathbb{R} \text{ nerazcepni člen}}$$

$$\downarrow$$

$$\frac{-b \pm \sqrt{D}}{2a}$$